

SmartHub

AI-Powered Personal Workspace

TKM Internship Programme • 2026

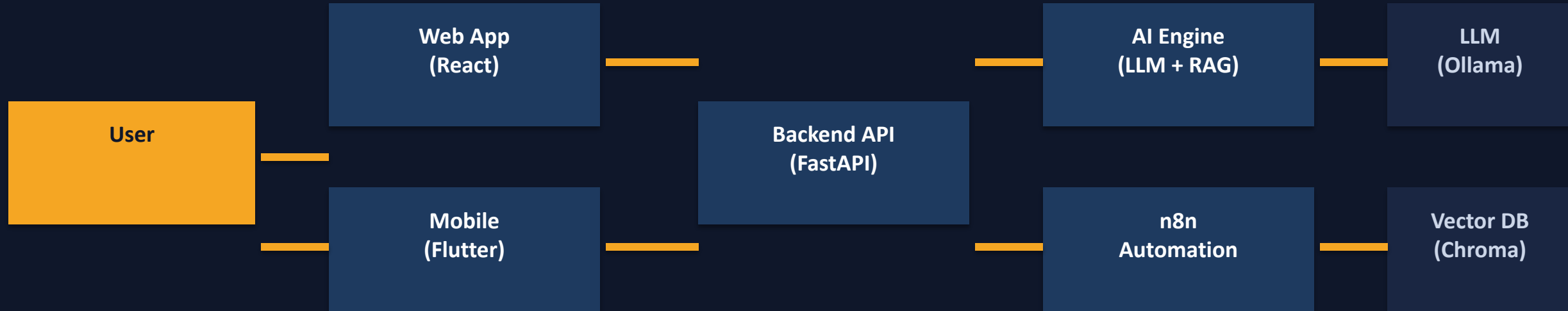
What is SmartHub?

- › Manage your workspace
 - Notes, tasks, documents, personal knowledge base
- › Chat with AI
 - Ollama / OpenAI / Gemini — your choice
- › Upload docs, ask questions (RAG)
 - Q&A from your own files
- › Smart notifications via n8n automation
- › Flutter mobile app synced with web app

Why This Project?

SmartHub Module	Maps to CIXIO Product
Backend API	cixio-mvp-claude-backend
Web App (React)	cixio-mvp-claude-webapp
Mobile App (Flutter)	cixio-mobile-app
AI/LLM Engine	cixio-llm, cixio-ai
n8n Automation	cixio-notifications
Docker + CI/CD	All CIXIO repos

SmartHub Architecture



Docker Compose wraps everything · GitHub Actions CI/CD · Prometheus + Grafana (G1–G3)

7 Modules — Every Group Builds All of Them

M1	Backend API	FastAPI / Node.js · JWT auth · CRUD · Swagger docs
M2	Web App	React / Next.js · Dashboard, Notes, AI Chat, Doc Upload
M3	Mobile App	Flutter · Login, Notes, AI Chat, Settings
M4	AI / LLM Engine	Chat completion · RAG pipeline · Vector DB (Chroma)
M5	n8n Automation	3 workflows: Welcome email · Daily digest · AI Insight
M6	DevOps / CI-CD	Docker Compose · GitHub Actions · Cloud deploy (bonus)
M7	Monitoring	Prometheus + Grafana · G1–G3 expected · All others: bonus

M1

Backend API

Owner: Backend students · Tech: Python FastAPI / Node.js Express

✓ Must Build (all groups)

- User auth (JWT register/login/me)
- CRUD: Notes, Tasks, Documents, Profile
- File upload endpoint + metadata in DB
- Health check · Swagger/OpenAPI docs
- SQLite (dev)

★ Advanced (G1–G3 expected)

- PostgreSQL · Pagination + filtering
- Rate limiting · Background tasks (n8n webhook)
- Unit tests — pytest/Jest $\geq 60\%$ coverage

✓ Must Build (all groups)

- Login / Register page
- Dashboard (notes list, tasks, recent docs)
- Notes: create, edit, delete (markdown/rich text)
- AI Chat: send message → response displayed
- Document upload + file list
- Responsive (works on mobile browser)

★ Advanced (G1–G3 expected)

- Dark mode · Real-time updates (WebSocket)
- RAG Query UI: upload doc → ask question → source citations
- PWA + service worker (offline support)
- Search across notes

M3

Mobile App (Flutter)

Owner: Mobile students (all groups contribute) · Tech: Flutter 3.x — Dart — required

✓ Must Build (all groups)

- Login / Register screen
- Home dashboard — notes list
- Notes: view, create, edit, delete
- AI Chat screen — send & display
- Settings screen (profile, logout)

★ Advanced (G1–G3 expected)

- Document upload from camera/gallery
- Push notifications (FCM)
- Offline mode + local SQLite sync
- Biometric login · RAG Query UI

M4

AI / LLM Engine

Owner: AI/LLM students · Tech: Python + FastAPI · Ollama / OpenAI / Gemini

✓ Must Build (all groups)

- POST /ai/chat — chat completion with history
- POST /ai/rag/ingest — upload & embed document
- POST /ai/rag/query — ask question, get answer + citations
- Vector DB: ChromaDB (recommended)
- Chunking: 500–1500 chars with overlap
- Embeddings: nomic-embed-text or OpenAI

★ Advanced (G1–G3 expected)

- Multi-document RAG · Reranker
- Conversation memory (multi-turn RAG)
- Token usage logging + cost estimation
- Support .docx and .pptx files

M5

n8n Automation

Owner: Assigned to AI or DevOps student · Tech: n8n (CIXIO-hosted) · Export as JSON

✓ Must Build (all groups)

- Workflow 1: Welcome email after registration
- Workflow 2: Daily task digest (9 AM cron)
- Workflow 3: AI insight report — summarise recent notes
- Export all as .json in /automation/ folder

★ Advanced (G1–G3 expected)

- Workflow 4: New document upload notification + AI summary
- Workflow 5: Overdue task reminder (hourly check)

M6

DevOps / CI-CD

Owner: DevOps student (or highest-GitHub backend student) · Tech: Docker · Docker Compose · GitHub Actions

✓ Must Build (all groups)

- Docker Compose: backend, frontend, ai-engine, db, ollama
- All services on private Docker network · .env for config
- Volumes for persistent data (DB, uploads, models)
- GitHub Actions CI: lint + test + Docker build on push

★ Advanced (G1–G3 expected)

- CD: auto-deploy to Railway / Render / Fly.io (free tier)
- Multi-stage Dockerfile (builder → minimal prod image)
- Push image to GitHub Container Registry (GHCR)
- HTTPS via cloud platform TLS

M7

Monitoring (Advanced)

Owner: G1–G3 expected · All others: bonus · Tech: Prometheus · Grafana · Docker Compose add-ons

✓ Must Build (all groups)

- Add prometheus + grafana to docker-compose.yml
- API request rate (req/min) · Error rate (4xx/5xx)
- AI chat usage (total chats/day)
- System health (CPU, memory)

★ Advanced (G1–G3 expected)

- Loki for log aggregation
- Grafana alerting rules
- Custom dashboard panels per group
- Export Grafana dashboard as JSON

8 Groups — 54 Students

Group	Size	Description	Target
G1 ALPHA	6	Elite — Startup-Ready	All 7 modules + cloud + monitoring
G2 BETA	7	Strong — High Potential	All 6 core modules + cloud deploy
G3 GAMMA	7	Mid-Upper — Developing Fast	All 6 core modules + Docker
G4 DELTA	7	Mid — Needs Guidance	Modules 1–6, quality over quantity
G5 EPSILON	7	Mid — Guided Learning	Core modules, strong AI focus
G6 ZETA	7	Mid-Low — Fundamentals	M1, M2, M4 well + basic M3 + n8n
G7 ETA	7	Building — Foundation Work	M1, M2, M4 first, M3/M6 secondary
G8 THETA	6	Entry — Intensive Mentoring	M1, M2 + basic AI chat + Workflow 1

Approved Tech Stack

Backend	Python 3.11 + FastAPI (recommended) · Node.js + Express (allowed)
Frontend	React 18 + Vite + Tailwind (recommended) · Next.js · Vue 3 (allowed)
Mobile	Flutter 3.x — required. React Native / Kotlin / Swift — NOT allowed
AI Engine	FastAPI + Ollama SDK · LangChain · OpenAI SDK · Gemini SDK
Vector DB	ChromaDB (recommended) · FAISS (allowed)
Automation	n8n (CIXIO-hosted instance) · GitHub Actions
DevOps	Docker + Docker Compose · GitHub Actions · Railway / Render / Fly.io

8-Week Timeline

Wk 1	Kickoff + Setup	Repo created, README, arch diagram, dev env running, roles assigned
Wk 2	Backend Core	Auth (JWT), CRUD for Notes. Initial Dockerfile
Wk 3	Frontend + Mobile Core	Web login + notes list. Flutter login screen. AI chat endpoint live
Wk 4	Integration Sprint 1	All modules talking. Web shows AI chat. n8n Workflow 1 working
Wk 5	AI/RAG + Automation	RAG pipeline working. n8n Workflows 2 & 3. Flutter Notes CRUD
Wk 6	Integration Sprint 2	Full Docker Compose. GitHub Actions CI passing. Mobile complete
Wk 7	Polish + Advanced	Bug fixes. Unit tests. Advanced features. Cloud deploy attempt
Wk 8	Final Demo + Submission	Live demo. All deliverables submitted. Code freeze

★ Week 4: Mid-point live demo · ★ Week 8: Final 45-min demo + Q&A · G8: Extra weekly check-in every week

What You Submit — Week 8

A	GitHub Repository	Public repo. smarthub-[groupname]. Clean commits. No secrets. .gitignore set up.
B	README.md	Arch diagram, tech stack, how to run, API docs link, team roles, demo video link.
C	Working Application	docker-compose up must start the full stack. All core features working.
D	n8n Workflow Files	All workflows exported as .json in /automation/ folder. README for each.
E	Demo Video	5–10 min screen recording. Login → Note → Upload → AI Q&A → n8n workflow.
F	Architecture Diagram	Mermaid or PNG. Shows user → web/mobile → backend → AI → vector DB → LLM.

Evaluation — How You're Scored

Functionality	30%	Core features work. App doesn't crash. API returns correct responses.
Code Quality	20%	Readable. No duplication. Small functions. No hardcoded secrets.
Integration	20%	All modules connected. Docker Compose brings up the full stack.
AI Implementation	15%	Chat works. RAG returns relevant answers with source citations.
Documentation	10%	Complete README, arch diagram, clear demo video.
Bonus: Advanced	5%	Cloud deploy, monitoring, PWA, push notifications, etc.

Rules & Boundaries

✗ Not Allowed

- Access CIXIO codebase or GitHub Enterprise
- Access CIXIO AWS / GCP / Azure accounts
- Store CIXIO customer or real personal data
- Commit API keys, passwords, or tokens — ever
- Share CIXIO's n8n credentials with anyone

✓ Allowed

- Use free-tier LLMs: OpenAI, Gemini, Ollama
- Use free cloud: Railway, Render, Fly.io, Supabase
- Use any open-source libraries (npm, PyPI, pub.dev)
- Ask your CIXIO mentor technical questions
- Reference CIXIO public products for inspiration

What CIXIO Gets — And What You Get

CIXIO gets:

- › 8 independent versions of a real AI app
- › Flutter screens potentially for cixio-mobile-app
- › n8n workflows — best ones imported directly
- › RAG pipeline ideas benchmarked against cixio-ai
- › Docker + CI/CD templates reused across repos
- › Talent pipeline — identify who's ready for full-time

You get:

- › Written feedback on every module
- › Letter of recommendation — top G1/G2/G3 completions
 - › Credit in CIXIO CONTRIBUTORS.md if code is integrated
- › Interview opportunity for top 3–5 individuals
- › Real production-grade project in your portfolio
- › 8 weeks of building something that actually matters

Let's build something real.

*8 groups. 54 students. 1 project. 8 weeks.
Your code might end up in a real product.*